WALCHAND COLLEGE OF ENGINEERING, SANGLI

(Government Aided Autonomous Institute)

WALCHAND LINUX USERS' GROUP

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METAMORPHOSIS

LIMITED SEATS

SAILING THROUGH THE CONTAINERS ERA

EXCITING PRIZES

11 & 12 APRIL

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- GO KICKSTART
- GO UNLEASHED

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TABLE OF CONTENT

- Goroutines
- Channels
- APIs
- Standard Libraries
- Json Encoding and Decoding
- Http Server

GOROUTINES

- Used to handle concurrent Programmes
- Goroutines can be thought of as lightweight threads
- The cost of creating a Goroutine is tiny when compared to a thread. Hence Go applications mainly have thousands of Goroutines running concurrently

GOROUTINES

Syntax:

- Put a keyword “go” before calling the name of Function

GOROUTINES

WITH goroutines:Runs at the same time

```go
func task1() {
    fmt.Println("Task 1")
}
func task2() {
    fmt.Println("Task 2")
}

func main() {
    task1()
    task2()
}
/*Output:
Task 1
Task 2
```

GOROUTINES

WITH goroutines:Runs at the same time

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/*Output:
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```

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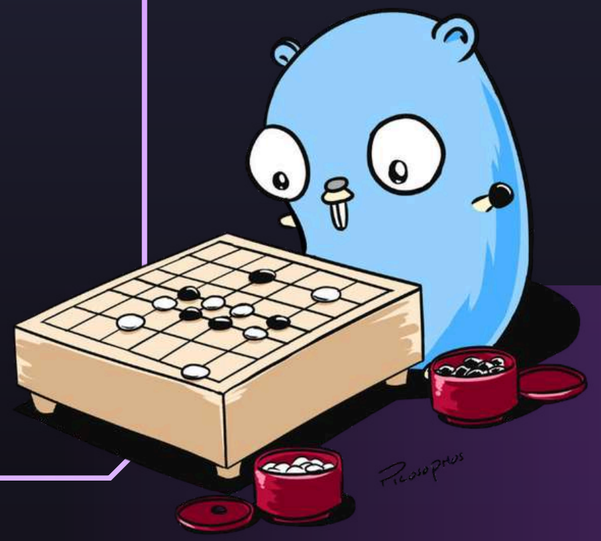
CHANNELS

CHANNELS

- Pipes using which Goroutines communicate
- Allows us to move data between different goroutines
- Do not communicate by sharing memory, instead share memory by communicating



SYNTAX

```go
chan T


a:=make(chan int)
```

channel of type T

Defines an int channel

SENDING, RECEIVING DATA

```
data := <- a // read from channel a

a <- data // write to channel a
```

- The direction of the arrow with respect to the channel specifies whether the data is sent or received

CHANNELS

- Sends and receives are blocking by default
- A blocking statement is a statement which blocks the execution of further lines of code after it

DEADLOCK

- If a Goroutine is sending data on a channel, then it is expected that some other Goroutine should be receiving the data

Interviewer
Explain deadlock and I'll hire you
Hire me and I'll explain it to you
Programmer

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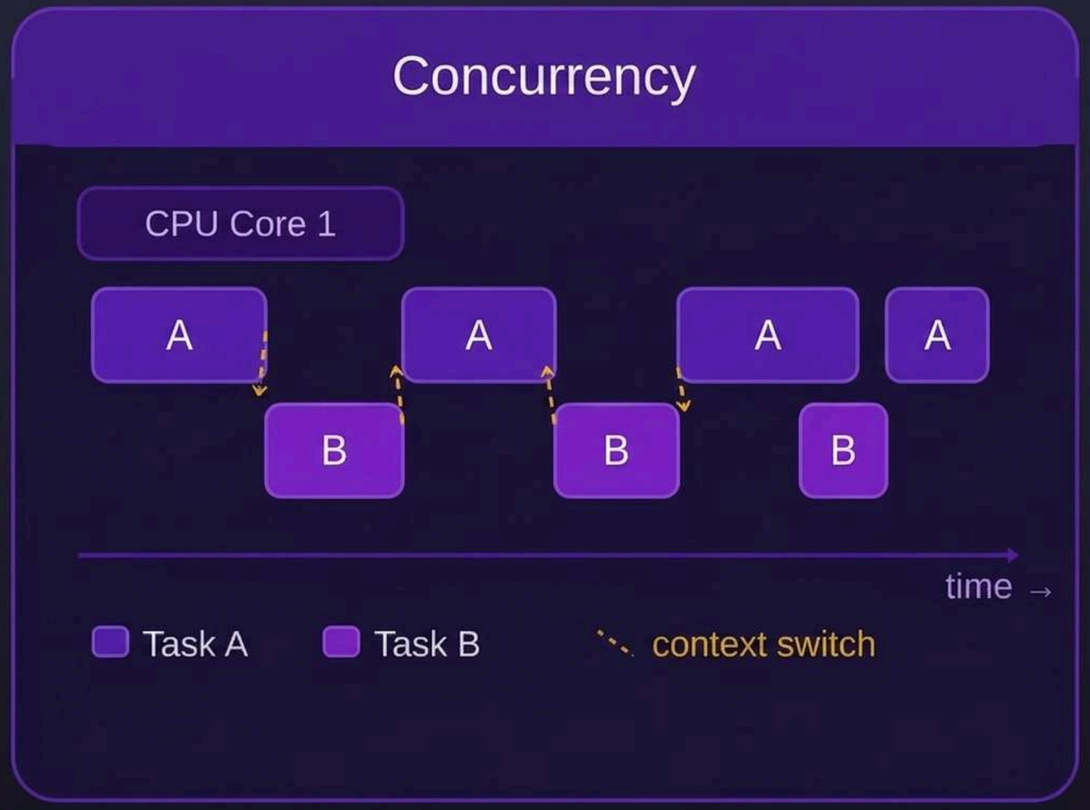
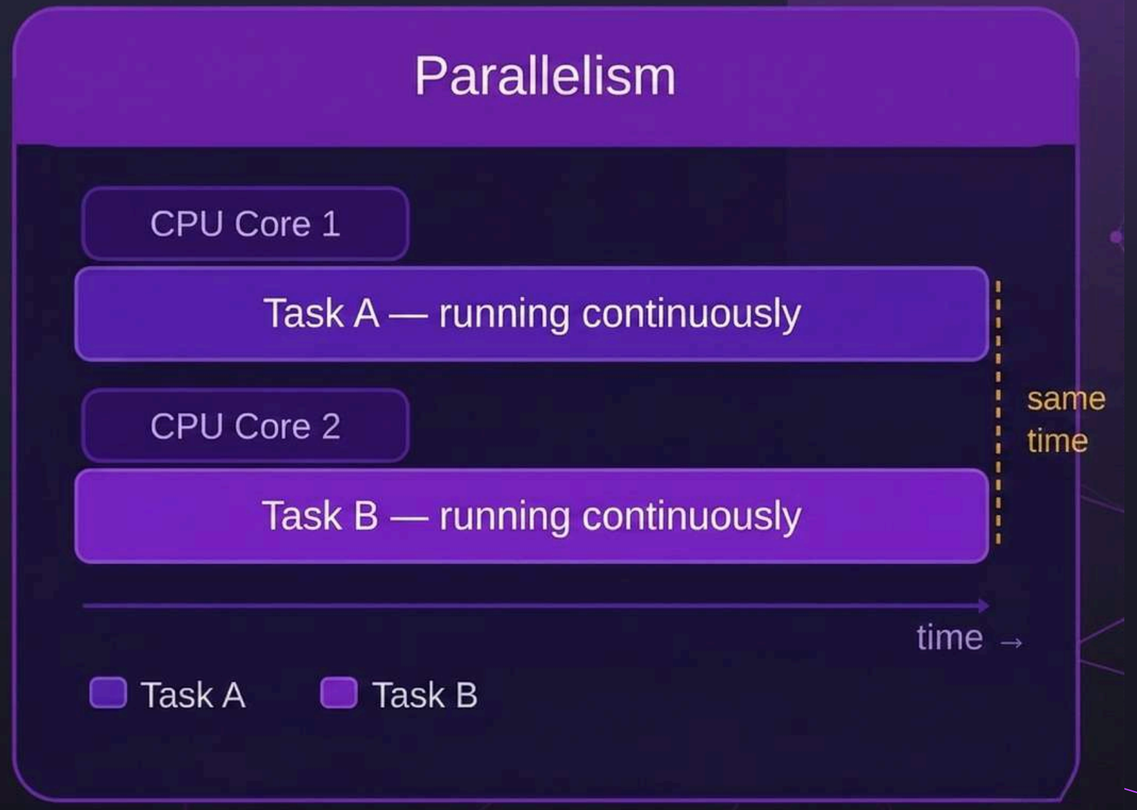
CONCURRENCY VS PARALLELISM


Concurrency
CPU Core 1
A
B
A
B
A
B
A
time →
Task A
Task B
context switch


Parallelism
CPU Core 1
Task A — running continuously
CPU Core 2
Task B — running continuously
same time
time →
Task A
Task B

CONCURRENCY

- Concurrency involves structuring a program so that two or more tasks may be in progress simultaneously

PARALLELISM

- Parallelism allows for two or more tasks to be executed simultaneously
- Parallelism requires more than one processor or thread, concurrency does not

APPLICATION PROGRAM INTERFACE

DEFINITION OF API

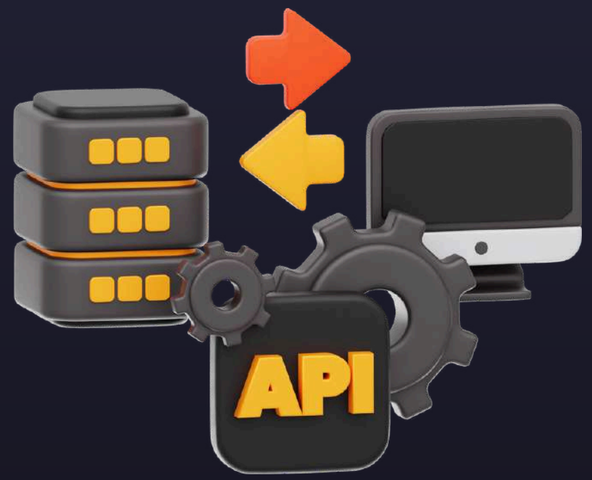
- Application Programming Interface (API) is a software interface that allows two applications to interact with each other

TYPES OF API

- Open APIs
- Internal APIs
- Partner APIs
- Composite APIs

API ARCHITECTURE

- REST
- SOAP
- Graph QL



HTTP METHODS

- GET
- PUT
- UPDATE
- DELETE

DATA FORMATS

- JSON
- XML
- Plain text

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CLIENT
HOW DO I GET THIS DATA?
SERVER
THAT'S THE NEAT PART
405
You wouldn't get it

STANDARD LIBRARIES USED

FMT

Library	Overview
“fmt”	Package fmt implements formatted I/O with functions analogous to C's printf and scanf.

NET/HTTP

Library	Overview
"net/http"	Package http provides HTTP client and server implementations

TIME

Library	Overview
“time”	Package time provides functionality for measuring and displaying time

ENCODING/JSON

Library	Overview
"encoding/json"	Package json implements encoding and decoding of JSON

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JSON ENCODING

JSON ENCODING

- Most web APIs return JSON responses
- encoding/json package
- NewEncoder
- Marshalling
- json tags

WHAT IS MARSHALLING?

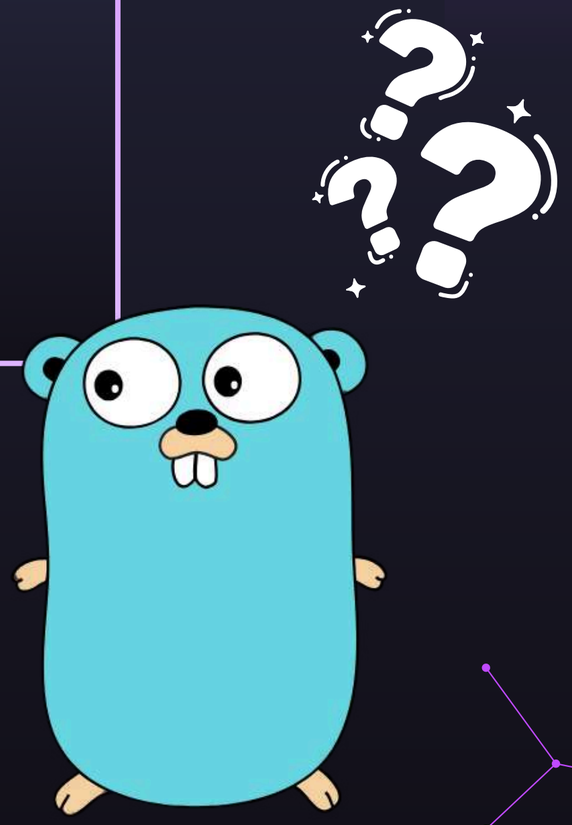
- Converting Go struct → JSON
- Key Function : json.Marshal()

WHAT IS NEWENCODER?

- Used for streaming JSON directly to output
- Common in HTTP responses
- Key Function : json.NewEncoder()

WHY USE NEWENCODER?

- Avoids intermediate byte array
- More efficient for large data
- Cleaner for web servers



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HTTP SERVER

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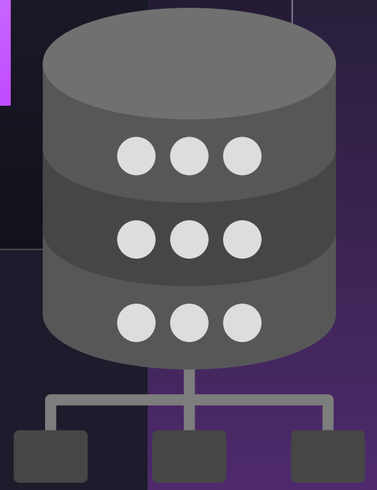
WHAT IS SERVER?



SERVER

- A server is simply a program or machine that provides services to other programs or users

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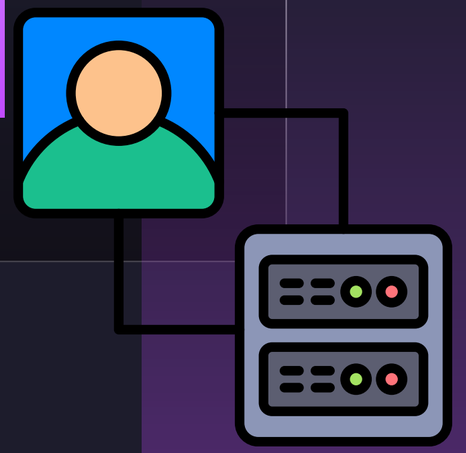
WHAT IS CLIENT?



CLIENT

- Client is anything that sends a request to a server

HTTP

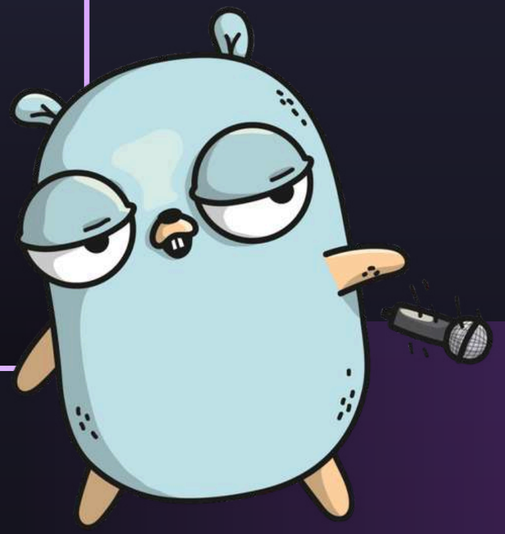
- HTTP stands for HyperText Transfer Protocol
- Request contains:
  - Method (GET, POST)
  - URL
  - Headers
  - Body
- Response contains:
  - Status code
  - Headers
  - Data

HTTP SERVER

- An HTTP server listens for requests from clients (like browsers) and sends responses.
- Common tasks:
  - Receive request from browser
  - Process request
  - Send response (HTML, text, JSON, etc.)

WHY GO FOR HTTP SERVERS?

- Built-in net/http package
- Very fast and lightweight
- Simple syntax
- Good for backend APIs and microservices



CREATING A BASIC HTTP SERVER

- Create handler function
- Register route
- Start server

NET/HTTP PACKAGE

- Go provides the net/http package to build web servers easily.
- Main components:
  - http.HandleFunc() → Register route
  - http.ListenAndServe() → Start server
  - http.ResponseWriter → Send response
  - http.Request → Access request data

HTTP METHOD

- HTTP methods define the type of action a client wants to perform on a resource.

GET	Retrieve data
POST	Create new data
PUT	Update existing data
DELETE	Remove data

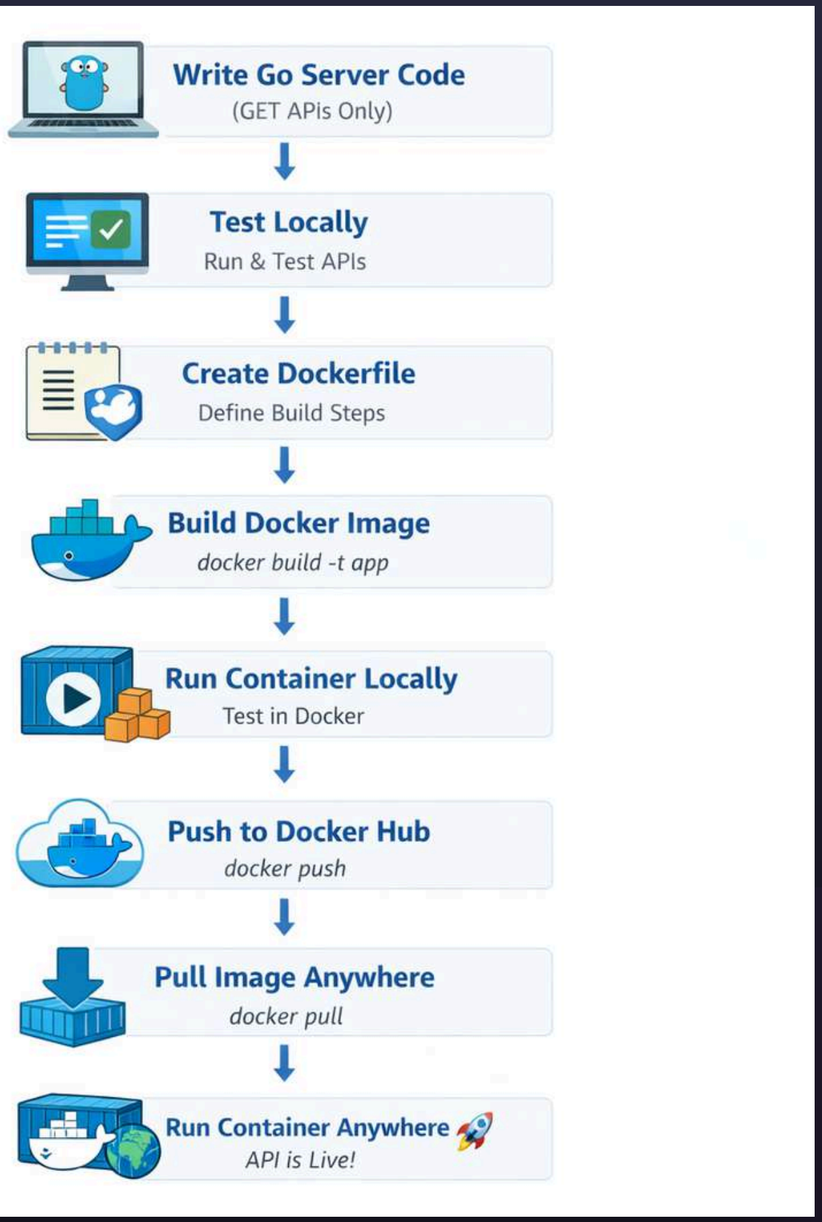
POSTMAN

- API testing tool
- Helps send HTTP requests easily
- No frontend required

POSTMAN

POSTMAN INSTALLATION

- Download Postman


Write Go Server Code
(GET APIs Only)
Test Locally
Run & Test APIs
Create Dockerfile
Define Build Steps
Build Docker Image
docker build -t app
Run Container Locally
Test in Docker
Push to Docker Hub
docker push
Pull Image Anywhere
docker pull
Run Container Anywhere
API is Live!

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THANK YOU